Arithmetic Sequences as Linear Functions
Answer Key
Algebra 1

Quick Answers

1. 40	4. 91	7. 900	10. 585
2. -8	5. 8	8. 71	
3. 181	6. 120	9. 6, 11	

Curriculum

Level: Algebra 1
HSF-BF.A.2, HSF-LE.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

2. *If they got 104: added the common difference instead of subtracting it, treating a decreasing sequence as increasing*
3. *If they got 190: multiplied by the number of chairs instead of by one fewer, adding a twelfth rise that is not there*
4. *If they got 97: counted 15 steps from the first term instead of 14*
6. *If they got 135: started counting at the second hour, so only seven pumping hours were subtracted*
7. *If they got 840: dropped the joining fee, using only the monthly charge*
10. *If they got 270: treated every row as the first row, so the extra three seats per row were never added*

1. Tiling pattern 5, 12, 19, 26, . . .

Step 1 (find d). Subtract consecutive terms: $12 - 5 = 7$, and $19 - 12 = 7$ as well, so the step is constant and the sequence is arithmetic with $d = 7$. **Step 2 (write both lines).** $a_1 = 5$ and $a_n = a_{n-1} + 7$ for $n \geq 2$. Both lines are required — the step alone does not fix the sequence. **Step 3 (count the steps).** Stage 6 is 5 steps past stage 1, so $a_6 = 5 + 5(7) = \boxed{40}$ tiles.

2. Drone altitudes 48, 41, 34, 27, . . .

Step 1 (find d). $41 - 48 = -7$, so the sequence decreases by a constant 7 and $d = -7$. Writing d with its sign is what keeps the rest of the work honest. **Step 2 (write both lines).** $a_1 = 48$ and $a_n = a_{n-1} - 7$ for $n \geq 2$. **Step 3 (count the steps).** The 9th reading is 8 steps past the first: $a_9 = 48 + 8(-7) = 48 - 56 = \boxed{-8}$. A negative altitude means the drone has passed below the height that was called zero — the launch deck — by 8 meters.

3. Stack of chairs: 82 cm for one chair, 9 cm per extra chair.
Step 1 (identify the two pieces). The first chair gives the starting height and each later chair contributes the constant rise, so $a_1 = 82$ and $d = 9$. **Step 2 (write both lines).** $a_1 = 82$ and $a_n = a_{n-1} + 9$ for $n \geq 2$, with a_n measured in centimeters. **Step 3 (count the steps).** A stack of 12 chairs is 11 rises past the first chair, not 12: $a_{12} = 82 + 11(9) = 82 + 99$, so the stack stands $\boxed{181 \text{ cm}}$ tall.

4. $a_1 = 7$**,** $a_n = a_{n-1} + 6$**; find** a_{15}**.**
Step 1 (read the rule). The rule adds 6 each time it is applied, and it starts from 7. **Step 2 (count the applications).** Going from a_1 to a_{15} applies the step 14 times, one for each arrow between the 15 terms. **Step 3 (compute).** $a_{15} = 7 + 14(6) = 7 + 84 = \boxed{91}$. Using 15 steps would answer a different question — it would give a_{16}.

5. $a_1 = 100$**,** $a_n = a_{n-1} - 12$**; which term equals 16?**
Step 1 (write the explicit form). After $n-1$ steps of -12 the value is $100 - 12(n-1)$. **Step 2 (set up the equation).** The question asks which n makes that value 16, so solve $100 - 12(n-1) = 16$. **Step 3 (solve).** $100 - 12n + 12 = 16$, so $112 - 12n = 16$, giving $12n = 96$ and $n = \boxed{8}$. Check: $100 - 12(7) = 16$ ✓.

6. Tank: 240 L, pump removes 15 L at the end of each hour.
Step 1 (fix the index). The 240 L is the volume *before* any pumping, so it is not a_1. At the end of hour 1 the tank holds $240 - 15 = 225$ L, so $a_1 = 225$. **Step 2 (write both lines).** $a_1 = 225$ and $a_n = a_{n-1} - 15$ for $n \geq 2$, with a_n in liters. **Step 3 (compute).** At the end of hour 8 the pump has run 8 times: $240 - 8(15) = 240 - 120$, so $\boxed{120 \text{ L}}$ remain. Equivalently $a_8 = 225 + 7(-15) = 120$, which is the same count of pumping hours reached from a_1.

7. Gym: 60-dollar joining fee, then 35 dollars a month.
Step 1 (write the recursion). The total after one month is $60 + 35 = 95$ dollars, and each further month adds 35, so $a_1 = 95$ and $a_n = a_{n-1} + 35$ for $n \geq 2$. **Step 2 (convert to explicit).** A recursive rule with constant step d is a linear function: the slope is $d = 35$ and the value at $n = 0$ is $a_1 - d = 95 - 35 = 60$, the joining fee. So $a_n = 35n + 60$. **Step 3 (evaluate).** After 24 months, $a_{24} = 35(24) + 60 = 840 + 60$, so the member has paid $\boxed{900}$ dollars. Dropping the fee would give only 840 dollars.

8. $a_1 = -5$**,** $a_n = a_{n-1} + 4$**; explicit rule and** a_{20}**.**
Step 1 (name the pieces). The constant step is the slope, $d = 4$, and the starting value is $a_1 = -5$. **Step 2 (write the explicit rule).** $a_n = a_1 + (n-1)d = -5 + 4(n-1)$, which simplifies to $a_n = 4n - 9$. The slope is 4 and the value at $n = 0$ is -9 — one step back from a_1, which is why $a_1 - d$ is the intercept and a_1 is not. **Step 3 (evaluate).** $a_{20} = 4(20) - 9 = 80 - 9 = \boxed{71}$; the recursive count agrees: $-5 + 19(4) = 71$.

9. Table: $a_3 = 23$ **and** $a_7 = 47$**.**
Step 1 (count the gap). From term 3 to term 7 is $7 - 3 = 4$ steps, and the value rises by $47 - 23 = 24$. **Step 2 (find d).** Each step is the same, so $d = \dfrac{47 - 23}{7 - 3} = \dfrac{24}{4} = \boxed{6}$. This is the slope of the line through the points $(3, 23)$ and $(7, 47)$. **Step 3 (work back to a_1).** Term 3 is 2 steps past term 1, so $a_1 = 23 - 2(6) = \boxed{11}$, and the recursive formula is $a_1 = 11$ with $a_n = a_{n-1} + 6$ for $n \geq 2$. Check: $11, 17, 23, \ldots$ reaches 23 at term 3 ✓.

10. Theater: row 1 has 18 seats, each row adds 3.

Step 1 (write the recursion). $a_1 = 18$ and $a_n = a_{n-1} + 3$ for $n \geq 2$; explicitly, $a_n = 18 + 3(n-1) = 3n + 15$. **Step 2 (find the last row).** Row 15 has $a_{15} = 18 + 14(3) = 60$ seats. **Step 3 (add the rows).** Pair the first row with the last, the second with the second-to-last, and so on: each pair totals $18 + 60 = 78$, and there are $15/2$ such pairs, so the sum is $\frac{15}{2}(18 + 60) = \boxed{585}$ seats. That pairing argument is exactly the "first plus last, times half the number of rows" shortcut: the amount one row gains, its partner loses, so every pair has the same total.